Predicting car safety-feature bundle choices

Discrete-choice structure, observed heterogeneity, and robust validation

Imelda Lee Woon Zee Ning Clarence Elvareta Sng Zhenhao

Note

This report was generated with AI assistance under human direction. The analysis, model selection, and numerical results were produced and checked by the project team; the final text should be reviewed by the team before submission.

1 Problem, data, and validation

The task is to predict which of four car safety-feature bundles a respondent chooses. Alternatives 1–3 contain 19 safety-attribute level codes and a price; alternative 4 is a designed no-purchase option with every attribute and price fixed at zero. The training data contain 21,565 choice tasks from 1,135 respondents, each completing exactly 19 tasks. The test data contain 4,997 tasks from 263 entirely new respondents. The outcome is evaluated by four-class log loss,

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{4} y_{ij}\log(p_{ij}),$$

for which uniform probabilities score 1.38629. The no-purchase option is chosen 30.2% of the time, rising from about 24% at Task 1 to 34% at Tasks 15–19. Consequently, both the inside-bundle comparison and survey fatigue matter.

The separation of respondents is the central validation constraint. Cases 1–1,135 occur only in train and Cases 1,136–1,398 only in test; therefore, random effects or memorized respondent preferences cannot transfer. A row-level split would also leak up to 18 tasks from the same person into the validation training set. We instead used an 80/20 respondent holdout (seed 7402) for rapid screening and respondent-grouped five-fold cross-validation (seed 4821) for confirmation. All reported cross-validation (CV) scores pool the out-of-fold predictions before calculating log loss. Model improvements were judged by paired respondent bootstrap comparisons, whose standard deviation was about 0.001, rather than by coefficient significance alone.

2 Best model: ensemble_v11 + neural ensemble diversity + set-context network + segment-shift pruning

Our best submitted model adds a set-context feed-forward network on top of the earlier **ensemble_v11 + MLP**, then prunes two structurally unreliable terms from its logit component:

$$\hat{\mathbf{p}} = 0.889\,\hat{\mathbf{p}}_{\text{v11+mlp}} + 0.111\,\hat{\mathbf{p}}_{\text{setctx}}, \qquad \hat{\mathbf{p}}_{\text{v11+mlp}} = 0.85\,\hat{\mathbf{p}}_{\text{v11}} + 0.15\,\hat{\mathbf{p}}_{\text{mlp}}, \qquad \hat{\mathbf{p}}_{\text{v11}} = 0.80\,\hat{\mathbf{p}}_{\text{m8trpg}} + 0.20\,\hat{\mathbf{p}}_{\text{xgb}}.$$

The set-context addition achieves **1.143533 respondent-grouped CV log loss and 1.200 public leaderboard log loss**, improving on both counts over ensemble_v11+MLP (1.143789 CV / 1.201 public) and substantially over the 1.38629 benchmark. Its canonical-split confidence interval was a wide near-miss, but six additional independent fold-seed assignments (repeated cross-validation) strengthened the signal rather than weakening it–the reverse of this project's usual near-miss pattern–and cleared our pre-stated significance bar. This is the second refinement, out of many dozens tested after ensemble_v11, whose public-leaderboard result was checked against its CV-predicted direction and matched (Section 3.3).

A final refinement, **segment-shift pruning**, then improved `m8trpg` itself. A structural audit (Section 3) found that two of its already-deployed market-segment terms (`In_seg3`, `In_seg5`, capturing segments 3 and 5's opt-out-margin deviation) are statistically indistinguishable from zero on the full training set (p=0.758, p=0.404) even though their *price-sensitivity* counterparts are highly significant (p<1e-10)–and segments 3 and 5 are exactly the segments whose share of the loss function explodes from 9.4% in training to 68.8% in test. Dropping only these two terms improved respondent-grouped CV, propagated through the full ensemble, from **1.143533 to 1.143255** (repeated CV: 6/6 seeds positive, 95% CI [0.0000441, 0.0003292], excluding zero). This is our final selected Kaggle submission: **1.200 public, 1.199 private** leaderboard log loss, now that the competition has closed and the private score is visible (Section 3).

2.1 Structured conditional-logit component

The main component, **m8trpg**, is a conditional logit estimated with `mlogit` (Croissant 2020). Alternative 4 has utility fixed at zero. For each inside alternative, utility contains:

- dummy-coded levels for all 19 safety attributes, avoiding an unjustified linear-in-code assumption;
- a saturated main effect for price levels 2–12;
- indicators for being cheapest or dearest within the three-bundle choice set, plus distances from the cheapest and dearest prices;
- alternative dummies for positions 2 and 3;
- linear price and inside-good interactions with observed respondent heterogeneity: income, age, mileage, night driving, market segment, task position, region, and parking availability, with selected inside-good interactions for gender, urbanicity, and education.

Respondent covariates cannot enter conditional logit as uninteracted main effects because they are constant across the four alternatives and cancel from the choice probability. Interacting them with price or an inside-good indicator lets observed heterogeneity transfer to unseen respondents. Task-position terms capture the increasing no-purchase rate across the 19-task survey. On its own, m8trpg scores **1.147021 CV**.

2.1.1 Price identification and choice-set context

Price produced the most important late-stage modeling advance. Earlier models treated it as one continuous slope even though every safety attribute was dummy-coded. A full price factor initially caused an exactly singular Hessian. This was not a numerical optimizer failure: design-matrix QR diagnostics exposed a structural rank-one dependency. Every inside bundle has exactly 9 of its 19 attributes at a non-reference level, so

$$\frac{1}{9}\sum_{k=1}^{19} I(\text{attribute } k \neq 0) = I(\text{inside bundle}) = \sum_{\ell=1}^{12} I(\text{Price} = \ell).$$

Thus, the attribute block and all twelve positive-price dummies independently reconstructed the same inside indicator. We resolved the redundancy by using price levels 2–12 and treating level 1 as the omitted price category; Price 0 for the opt-out shares that reference cell because inside-versus-opt-out is already represented by the attribute block. This loses no identified information. The fitted level effects were monotone and convex (about -0.66 at level 2 to -3.64 at level 12), showing that the old linear-price specification had missed real curvature.

Choice-set context added complementary signal. `is_cheapest` and `is_dearest` encode price rank, while `price_gap_min` and `price_gap_max` encode the magnitude of the differences. Adding the price factor and rank terms improved CV from 1.15671 to 1.15160; the two gap terms then improved it to 1.147021. The gap improvement was about 0.0046, several paired-bootstrap standard deviations, and was achieved with only two additional parameters.

2.2 Diverse xgboost component

The second component is a wide-format `multi:softprob` xgboost model with 73 boosting rounds (Chen and Guestrin 2016). It is weaker alone (**1.178668 CV**) because it must learn the within-task choice structure from raw columns, but its errors differ from the conditional logit's. Out-of-fold weight search assigns it 20% and reduces CV log loss by 0.001927. The gain is small but directionally stable: the ensemble beat m8trpg in 96.6% of paired respondent bootstrap samples.

2.3 Neural ensemble diversity

Every flexible learner tried up to this point–CART, LASSO multinomial regression, and multiple xgboost variants–remained tree-based and never beat the structured logit alone. A small single-hidden-layer feed-forward network (eight hidden units, weight decay 0.1, base R's `nnet` (Venables and Ripley 2002)), trained on the same wide-format attribute/price/covariate representation, scores **1.190543 CV alone**–worse than either existing component–yet contributes a genuinely different error pattern: fold-cross-fitted blend weight 13–17%, and a respondent-bootstrap 95% interval for its ensemble contribution of [**+0.000092, +0.002513**], excluding zero at our pre-stated ordinary significance threshold (though not under a stricter correction for the architecture search that selected it). It was the only candidate, out of several dozen tested after ensemble_v11 (deeper networks, gradient-boosting variants with native categorical handling, distribution-shift corrections, higher-order interactions, and design-structure corrections described below), to clear this bar, and it was submitted for exactly that reason.

2.4 Set-context network

The final component learns, rather than hand-builds, choice-set context. Every model above encodes each alternative's own attributes and price plus a handful of hand-designed context features (`is_cheapest`, `price_gap_min/max`). A feed-forward network instead takes, for each alternative, its own features together with a permutation-invariant summary of the *other* alternatives in the same task (Zaheer et al. 2017)–in the spirit of set-pooling architectures for inputs with no natural ordering. Scoring **1.261810 CV alone**, it is the weakest standalone component yet contributes real, confirmed ensemble value via the same weak-alone-but-diverse pattern already established for xgboost and the shallow network. Its canonical five-fold CV gain was a wide, low-confidence near miss (mean **+0.000153**, 64% bootstrap win rate), triggering our pre-registered escalation to six independent fold-seed assignments; the pooled repeated-CV gain came back **larger**, not smaller (**+0.001164**, 95% interval [**+0.000011, +0.002318**], excluding zero, with all six repeats positive). Before submission, the full-data build was required to reproduce identically across two independent training runs (maximum difference **0.0**), guarding against any silent nondeterminism in the network fit.

Table 1 reports the main progression on comparable respondent-grouped validation. Single-split figures are shown only for early models that predated the canonical five-fold harness.

Table 1: Main model progression. Lower log loss is better.

Model	Main change	Local log loss	Public
Uniform benchmark	0.25 per alternative	1.3863	1.386
mod1	Linear conditional logit + ASCs	1.2357 (holdout)	1.270
mod2b	Factor-coded attributes; identified alt dummies	1.2186 (holdout)	–
mod6 / mod7	Price heterogeneity, then inside-good heterogeneity	1.2049 / 1.2024 (holdout)	– / 1.230
mod8	Segment-specific price and inside effects	1.1665 (CV)	–
m8tr	Task fatigue, region, parking	1.1567 (CV)	–
ensemble_v9	70% m8tr + 30% xgboost	1.1517 (CV)	1.204
m8trp	Saturated price + cheapest/dearest rank	1.1516 (CV)	–
m8trpg	Add price-gap magnitudes	1.1470 (CV)	1.213
ensemble_v11	80% m8trpg + 20% xgboost	1.1451 (CV)	1.202
ensemble_v11+MLP	+15% feed-forward net	1.1438 (CV)	1.201
+ set-context network	+11% set-pooling net	1.1435 (CV)	1.200
+ segment-shift pruning (final)	**Drop 2 unreliable segment terms**	**1.1433 (CV)**	**1.200 / 1.199 private**

The experiments also established useful negative results:

- **Alternative choice structures.** Nested logit did not improve the bundle-versus-opt-out split. Full and price-only mixed logits improved training fit but not prediction for unseen respondents. Two observed-membership latent-class extensions (McLachlan and Peel 2000) were also null. A task-fatigue mixture scored 1.147826 CV and its class effects changed sign across folds. A more central model applied a class-specific multiplicative scale to every price-related utility term. It found stable classes at about 0.5 and 1.85–2.0 times normal price sensitivity across all folds, but scored 1.147629 versus 1.147021 for m8trpg–a 0.0006 difference inside the noise floor. The discrete heterogeneity was real but redundant with continuous income, segment, and age interactions.

- **Flexible learners.** CART, LASSO multinomial regression, and xgboost alone remained behind the structured logit. A long-format binary xgboost followed by within-task renormalization scored 1.2053 versus 1.2042 for its wide multiclass counterpart; renormalization is not a genuine ranking objective. A subsequent `rank:ndcg` model grouped alternatives by choice task and improved xgboost's CV loss from 1.178668 to 1.163610, confirming that a true ranking loss learns useful within-task comparisons. A broader hyperparameter retune reached only 1.176029, however, and neither variant displaced ensemble_v11. A teammate's random forest reported 1.1616 under a row-level split but 1.259 publicly, illustrating the leakage caused by splitting repeated tasks rather than respondents.
- **Extra heterogeneity.** Attribute-by-segment terms, K-means personas, relative feature load, and additional price interactions with gender, urbanicity, or education did not generalize. Categorical price interactions with sparse night-driving and mileage bins caused quasi-separation; only balanced age bins gave a negligible gain. Extending cheapest/dearest rank flags to the other attributes was null because most level codes are unordered categories. Continuous three-way terms avoided the sparse-bin problem: Price $\times$ income $\times$ mileage had a negative coefficient in all five folds, but its ensemble gain was only 0.000495 and its respondent-bootstrap 95% interval [**-0.000689, 0.001660**] crossed zero. Segment-specific mileage-price slopes were significantly harmful. A log-income transformation was also null.
- **Combination methods.** A nested-CV conditional-logit stack on the base models' log probabilities (Wolpert 1992) scored 1.146126, marginally worse than the simple probability blend. With only two base models and a small minority weight on xgboost, the extra combiner flexibility added little. This conclusion held even after adding four more diverse components (below): a nested ridge-regularized log-linear opinion pool and a shallow xgboost meta-model both failed to beat simple arithmetic weighting a second time.

After ensemble_v11, a further, larger round of experiments pursued public leaderboard leaders' scores (reported near 1.186–1.190, against our 1.202). None closed that gap, but several are worth recording as a well-verified negative result rather than an unexamined ceiling:

- **Genuine ranking and regularized objectives.** A ranking-loss xgboost (`rank:ndcg` with task-level grouping), a re-tuned wide xgboost, and a regularized conditional logit fit via the stratified-Cox equivalence each modestly improved their own component (CV 1.1636, 1.1760, and 1.1643 respectively) and received stable minority weight in a combined blend, but an eight-component arithmetic blend of every model tried–including the neural network–reached only 1.142112 CV. Its bootstrap interval barely cleared zero on the canonical split but, when re-tested with five additional independent fold assignments (repeated cross-validation, chosen specifically to distinguish a real gain from a favorable split), the pooled interval fell back across zero–a fold-assignment artifact, not a genuine improvement.
- **Isolating and closing the two remaining threads.** The closest unconfirmed lead, a prior-smoothed price-reference feature bundled with a noisier attribute-familiarity term, was re-tested with the noisy term removed entirely (pre-registered, repeated cross-validation from the start): the isolated price-anchoring effect is real and directionally stable (its sign was negative in all 30 independent fold fits) but its size (about 0.0006) still sits inside the respondent-level noise floor. Separately, a weight-shared neural network trained on the exact four-way choice likelihood–the one untested combination of an exchangeable-alternative constraint with a flexible function class–screened positively but its canonical-CV gain shrank eight-fold versus the screen and its interval crossed zero comfortably. Two different function classes (tree and neural) now agree that the exact choice-likelihood objective was never the missing lever; the hand-built interaction structure is what does the real work.
- **Distribution-shift correction by retraining, not just reweighted evaluation.** A genuine weighted-likelihood refit toward test-like respondents (density-ratio weights, following Shimodaira (2000)) was decisively harmful (bootstrap interval excluding zero on the harmful side): down-weighting training respondents costs more effective sample size than it gains from targeting the shifted population.
- **Exact conditional-logit-consistent gradient boosting.** Wide multiclass and ranking-loss xgboost do not directly optimize the four-way softmax likelihood the way `mlogit` does. A custom xgboost objective enforcing the true multinomial gradient (verified against finite differences) failed cold-start, confirming that m8trpg's hand-built interaction structure, not the loss function, was doing the real work; used as a small residual correction on top of m8trpg's own fitted utility it gave a tiny, statistically unconfirmed gain.
- **A confirmed design-rotation structure.** Fingerprinting each respondent's complete ordered 19-task design sequence shows the survey reuses a pool of only **299 distinct questionnaire versions** across train and test combined, consistent with standard commercial CBC rotation defaults–a stronger, previously undetected version of the 98.5% single-task recurrence above. A cross-fitted, utility-space penalized correction to the opt-out option keyed by version was tested but rejected: each version has only 3–5 sharing respondents, too few to support even

one well-shrunk parameter without being dominated by a single respondent.

- **Two more tree-learner families and a pre-registered architecture search.** LightGBM (Ke et al. 2017) and CatBoost (Prokhorenkova et al. 2018), whose native categorical-splitting mechanisms differ from xgboost's, both received zero ensemble weight. A 24-configuration neural architecture search, with its multiplicity correction fixed *before* seeing any results, found a modestly better network but correctly rejected it under the pre-declared rule.
- **A further wave of four independently-verified experiments.** Learned choice-set-crowding features and a stagewise, component-wise boosting correction on m8trpg's fitted utility were both consistently-signed but tiny near misses that failed repeated cross-validation (gains of +0.000102 and +0.000038); a curvature-shrunk version of the saturated price factor was a clean, decisive reject (-0.0019), confirming that factor is not overfit; a calibrated radial-basis-function support vector machine–a materially different function class from every tree or neural component tried–scored worse alone than any other component and, unlike xgboost or the shallow network, made the ensemble worse rather than better.

These results show why selection used predictive loss rather than p-values: several added terms had very small p-values yet worsened the holdout score.

3 Public versus private leaderboard fit

The public leaderboard evaluates approximately 70% of the 4,997 test tasks; the remaining 30% formed the private leaderboard, unseen until the competition closed. The public gap has not been constant: mod1 was 0.034 worse than local validation, mod7 0.028 worse, ensemble_v9 0.052 worse, and ensemble_v11 0.057 worse. This prevented us from projecting the private score by simply adding a fixed correction to CV.

Now that the private leaderboard is visible, we can check this directly rather than only simulate it. Across 18 team submissions with both scores recorded, public and private log loss correlate at **r=0.979**; private was on average 0.004 *better* than public (sd 0.005, max difference 0.013)–the public leaderboard was, in practice, an unbiased, low-noise proxy for private here. CV predicted private slightly less tightly than public (r=0.787 vs. 0.845; mean gap 0.052 vs. 0.057), consistent with private being a smaller, noisier panel, as our bootstrap simulation anticipated. Ranking our own progression by private score alone does not perfectly reproduce the CV ranking–ensemble_v9 (1.152 CV, weakest of the group) edges out both set-context and segment-shift pruning (1.199 private each) at 1.197–but the spread across seven models is only 0.018, well inside the paired-noise band this report already quantifies: confirmation of, not an exception to, our own thesis that small private-rank differences here reflect noise, not quality.

3.1 Team submissions

For completeness, every teammate's independently submitted model is listed below, code-reviewed for validation soundness, not only the zhenhao-branch progression of Section 2.

Model	CV	Public	Private	Validation
Zeening: RF, grid-searched (`rf_gridsearch`)	1.162	1.259	1.253	row-level split–**leaky**
Zeening: RF+XGB+glmnet blend (`019_blended.R`)	–	1.216	1.226	respondent-grouped–sound
Imelda: 3-way MNL ensemble	1.228	1.263	1.254	respondent-grouped–sound
Imelda: MNL + xgboost	1.186	1.255	1.247	respondent-grouped–sound
Imelda: MNL + covariate interactions	–	1.249	1.245	respondent-grouped–sound
Imelda: as above, blended toward uniform	1.203	1.245	1.241	respondent-grouped–sound
Clarence: xgboost (task-based split)	–	1.221	1.216	task-based split–**leaky**

Model	CV	Public	Private	Validation
Clarence: ensemble, honest re-weighting	1.165	1.224	1.219	respondent-grouped–sound (our fix)

"–" marks a CV never captured. Two submissions were leaky at build time (Zeening's first: random row split; Clarence's: task-number split, not respondent) and had no trustworthy internal CV as a result; both were later fixed–we rebuilt Clarence's ensemble with honest fold-cross-fitted weights, and Zeening independently corrected her own methodology, which we verified implements respondent-grouped CV throughout. Every properly-validated teammate submission (five of eight) landed in a tight 1.19–1.26 band on both leaderboards with mlogit-family-sized gaps, reinforcing that respondent-grouped validation, not model family or author, is what separates a trustworthy estimate from a misleading one here. One further submission, `hierarchical_pool_v18_balanced` (public 1.200, private **1.194**, better than every model above), could not be traced to any script despite an exhaustive git-history search and is reported without further interpretation.

3.2 Distribution shift and experimental blocks

Adversarial validation using respondent covariates distinguished train from test at **AUC 0.634**, confirming covariate shift. Income was the dominant separator: median income was 60,000 in train and 80,000 in test, and some top brackets were more than three times as prevalent in test. However, this did not justify an unvalidated correction. Test-likeness weighting raised m8trpg's OOF loss from 1.1470 to about 1.1577, but the highest-income training tercile actually had the best loss. Two income-reweighting estimates put the shift's contribution to the 0.057 CV-public gap near 0.010 (about 18%), yet a respondent-bootstrap 95% interval of [**-0.0057, 0.029**] includes zero. We also tested a genuine density-ratio weighted refit, with both fitting and held-out evaluation targeted toward test-like respondents (Shimodaira 2000). Even the gentlest weight was harmful in every fold: target-weighted loss rose from 1.159787 to 1.161203, and the 95% interval for its gain was [**-0.002614, -0.000422**]. Reweighting reduced effective sample size enough that variance outweighed any targeting benefit. Log-income interactions did not improve validation. We therefore treat income shift as a real risk factor, not as a quantified causal explanation or a validated correction.

A second, more extreme shift was found by directly auditing the model's existing parameters rather than proposing a new one. The market-segment covariate shifts far more sharply than income: segment 6 is 27.0% of the 1,135 training respondents but 0% of the 263 test respondents, while segments 3 and 5 are only 9.4% of training combined but 68.8% of test–an adversarial-validation classifier on segment alone reaches **AUC 0.898**, dwarfing income's 0.634. A full-data coefficient audit of `m8trpg` found that segments 3 and 5's opt-out-margin deviation terms are statistically indistinguishable from zero even on the full training set (p=0.758, p=0.404) while their price-sensitivity counterparts are highly significant ($p<1e-10$)–two already-weak, already-deployed parameters whose loss-function weight explodes roughly sevenfold between training and test. Dropping only these two terms improved respondent-grouped CV on the whole population (not merely a subgroup), amplified monotonically on the 30% of training respondents most similar to test, and survived repeated CV at the full-ensemble level (Section 2)–the only case here where auditing an already-deployed parameter against a known shift, not a new mechanism, produced a promotable improvement.

The conjoint experiment is also heavily blocked: each task position draws from about 296 exact designs, seen by 3.8 respondents on average, and **98.5%** (4,921/4,997) of test tasks exactly repeat a train design. Cross-fitted empirical choice-share shrinkage improved 1.145094 only to 1.144742, within the paired noise floor; stronger use of cell frequencies hurt because three or four observations cannot estimate a reliable four-way share. The overlap is reassuring for design support but does not remove respondent-composition risk.

3.3 Why the ensemble should be retained

We directly tested whether xgboost caused a complexity or overfitting penalty by submitting m8trpg alone. Despite being simpler, it scored **1.213 public**, a 0.065979 CV-public gap–the largest among our trustworthy models–whereas ensemble_v11 scored 1.202 with a 0.056906 gap. Removing the weaker learner made the public result worse. The 20% xgboost component therefore acts as genuine variance reduction, not needless complexity. Adding the neural-network component (Section 2.3) then improved both CV (1.145094 to 1.143789) and public score (1.202 to 1.201) together–a 0.057211 gap, consistent with the established 0.052–0.057 pattern. At the time, this was the only refinement in the entire post-ensemble_v11 search whose public result was checked against its CV-predicted direction and matched–though, as the next submission below shows, not the last.

A second, independent submission reinforces this rather than merely repeating it. A candidate combining the neural network with an additional price-by-income-by-mileage interaction reached the best pre-submission CV number of any untested candidate at the time (1.143328) but its bootstrap interval against the current best already crossed zero; submitted anyway as the best available honest bet, it scored **1.210 public**–worse than the retained model, and outside a paired resampling simulation's predicted range. Its CV-public gap (0.0667) is the largest in the project, close to the plain-logit gap rather than the ensemble-class pattern.

A third submission then broke the losing streak. The set-context network (Section 2.4) cleared our bar via repeated cross-validation and was added to the ensemble at an 11.1% weight, frozen from the mean of six cross-fitted fold weights and locked behind a full-data build that was required to reproduce identically across two independent runs before submission. It scored **1.200 public**, improving on the prior best (1.201) with a 0.056467 CV-public gap–again inside the 0.052–0.057 ensemble-class pattern, not the wider gap seen for the two flexible-model submissions above. This is the second time a CV-predicted improvement direction has been confirmed on the public leaderboard, and together with the price-curve and interaction-augmented submissions it forms a consistent three-point pattern: well-behaved ensemble-class additions transfer reliably, while additions with unbounded or high-variance structure do not.

A fourth submission, segment-shift pruning, is a smaller and more targeted refinement than any of the above–a full-ensemble CV improvement of only 0.000278, an order of magnitude below the others. It scored **1.200 public**, identical to the prior submission at three-decimal display precision. Because the CV-predicted gain sits below both Kaggle's rounding granularity and this project's own quantified paired-comparison noise floor (about 0.001), this result neither confirms nor refutes the CV-predicted direction–unlike the three-point pattern above, it is genuinely uninformative at this resolution. We retain it as the standing best on the strength of its CV evidence (Section 2) and its structural rationale (a directly diagnosed distribution-shift mechanism, not a speculative new feature), not because the public leaderboard confirmed it.

Leaderboard differences must also be interpreted at the respondent level. A snapshot of the real public leaderboard (43 teams, 27 July 2026) placed us 16th of 43 at 1.201; the top 18 teams span only 0.016 of log loss, and 34 of 43 teams–including the leader–sit within one noise-standard-deviation of our score. Resampling public-sized sets of 184 whole respondents from OOF predictions gave a single-score standard deviation of **0.0225**, 2.14 times the naive row-level estimate of 0.0105. Among pairs of equally generated samples, **64.6%** differed by at least 0.015–comparable to the gap separating most of the leaderboard's middle tier. The field's compression is therefore not an artifact of our own simulation: it is visible directly in contemporaneous, independently-scored results, and is consistent with most leaderboard gaps near our position reflecting sampling noise rather than a missing modeling breakthrough. The smaller private panel will be at least as composition-sensitive. Our best defense is respondent-grouped CV, observed (transferable) heterogeneity, and a diverse ensemble–not tuning to one public split.

A final honest note on selection, made possible only after the leaderboard closed: one untracked submission scored better privately than our selected final (Section 3, Team submissions)–never a candidate at selection time, and, per the noise analysis above, not read as proof of a better model given the size of the gap relative to this section's own quantified noise band. We report it because a competition report should state the real outcome plainly rather than omit an inconvenient one.

4 Insights and limitations

Three behavioral insights were especially robust. First, price response is nonlinear and depends on both absolute price and its rank and distance within the displayed choice set. Second, observed respondent characteristics change price sensitivity and the propensity to purchase, while respondent-specific random effects do not transfer to new people. Third, task position matters: the rising opt-out share is consistent with fatigue or satisficing over the 19-choice sequence.

A fourth insight is structural rather than behavioral: the questionnaire itself is far more repetitive than any single respondent's 19 tasks suggest. Fingerprinting each respondent's complete ordered design sequence (train and test combined) shows the whole study draws from a shared pool of only **299 distinct questionnaire versions**, consistent with standard commercial conjoint-rotation practice. This explains the 98.5% single-task design recurrence as a property of the survey instrument rather than coincidence, but it does not translate into an exploitable feature: each version is shared by only 3–5 respondents, too few to support a version-specific parameter without that parameter being dominated by a single respondent's outcome.

The remaining error looks broadly stochastic rather than concentrated in a fixable subgroup. Across OOF predictions, observed and predicted frequencies agree within 1–2 percentage points over most of the 0–0.8 probability range. Although 49.5% of tasks are argmax misses, this is compatible with probabilistic calibration; 36.3% of misses have a probability gap above 0.30, but their rate is flat across segment, task bucket, region, and true class. The worst 10% of tasks contribute 21.4% of total loss, rather than dominating it. A flexible xgboost model also fails to uncover a large omitted nonlinear signal. These checks support–but cannot prove–that 1.143–1.145 CV is near the practical floor using the observed variables, a conclusion reinforced by an unusually large and diverse set of subsequent negative results (Section 2): alternative choice structures, distribution-shift correction by reweighting and by retraining, higher-order interactions, seed-bagging and cross-fitted shrinkage in both directions, an exact random-utility-consistent gradient-boosting objective, two further tree-learner families with different categorical-splitting mechanisms, a genuinely pre-registered architecture search, repeated cross-validation that overturned an apparent significant result rather than merely confirming ones already rejected, the newly confirmed design-rotation structure, a further wave of learned choice-set features, stagewise boosting, price-curve shrinkage, and a kernel-machine diversity component, and–in a final round–a dedicated opt-out/bundle hurdle model, four independent generalizations of the softmax link/scale (all rejecting decisively rather than merely falling short), a difficulty-predictability diagnostic (real but non-exploitable loss predictability), a set-context pooling-statistic addition, and six externally-proposed mechanisms spanning consideration-set screening, relative price-gap and salience weighting, and semantics-motivated taste heterogeneity and feature-family saturation, all converging on the same narrow ceiling. Four real submitted comparisons of a more flexible, differently-structured, or more rigorously-audited model against the plainer ensemble now exist: two favor the plainer ensemble (the standalone logit and the interaction-augmented candidate); the set-context network, a plain ensemble-class addition, confirmed a genuine, if small, improvement; and segment-shift pruning improved CV by an order of magnitude less, scoring identically to its predecessor at display precision–consistent with, not contradicting, the pattern. An independently-run adversarial search–explicitly instructed not to stop early–reached the same exhaustion conclusion via more than a dozen further genuinely new experiments, one of which (segment-shift pruning) was the only one to surface a promotable improvement, itself found not by proposing a new mechanism but by auditing an existing parameter against a known distribution shift.

Limitations remain. Stated choices contain unobserved taste and within-person inconsistency; train and test respondents differ measurably; absolute CV log-loss uncertainty is about 0.01 even though paired comparisons are much tighter; and the private split, though now scored, was smaller and noisier than the public one throughout model selection. Calibration and subgroup checks are descriptive, not guarantees under shift. Design-cell shrinkage, latent classes, and the tested ranking/regularization/reweighting/interaction variants do not exhaust all possible specifications or richer covariates.

References

Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–94. https://doi.org/10.1145/2939672.2939785.

Croissant, Yves. 2020. "Estimation of Random Utility Models in R: The mlogit Package." *Journal of Statistical Software* 95 (11): 1–41. https://doi.org/10.18637/jss.v095.i11.

Ke, Guolin, Qi Meng, Thomas Finley, et al. 2017. "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*.

McLachlan, Geoffrey J., and David Peel. 2000. *Finite Mixture Models*. Wiley. https://doi.org/10.1002/0471721182.

Prokhorenkova, Liudmila, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2018. "CatBoost: Unbiased Boosting with Categorical Features." *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*.

Shimodaira, Hidetoshi. 2000. "Improving Predictive Inference Under Covariate Shift by Weighting the Log-Likelihood Function." *Journal of Statistical Planning and Inference* 90 (2): 227–44. https://doi.org/10.1016/S0378-3758(00)00115-4.

Venables, William N., and Brian D. Ripley. 2002. *Modern Applied Statistics with S*. Fourth. Springer. https://doi.org/10.1007/978-0-387-21706-2.

Wolpert, David H. 1992. "Stacked Generalization." *Neural Networks* 5 (2): 241–59. https://doi.org/10.1016/S0893-6080(05)80023-1.

Zaheer, Manzil, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander J. Smola. 2017. "Deep Sets." *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*.